STUDY MATERIAL

Methods module

Comparing Europe with Data. Six methods explained with the COIL data, from the idea to the result.

Correlation · regression · panel data · classification · PCA · clustering · missing values

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HOW TO USE THIS MODULE

Four slides per method

1

The idea

What the method does, in plain words. No formulas you need to memorise.

2

A real example

Computed with the COIL dataset, so you can reproduce every number.

3

How to read it

What the output means, and the most common mistakes.

4

Do it yourself

Menus in Power BI and Gretl. Code in Python and R only if your course uses it.

You do not need to program. Everything in this module can be done with menus; the code is there for those who prefer it, and it is on the website ready to run.

THE WORKFLOW

Every project follows the same six steps



Three things that apply to every method

- The COIL data are 32 countries. That is a small sample: keep models simple and report n in every table.
- Cross-section = one year, many countries. Panel = many countries over many years. Choose the same year(s) for all countries.
- Association is not causation. Your results describe patterns; explain them with the national context, do not claim effects.

CHOOSING

Which method answers your question?

Your question	Method	Section
Do two indicators move together?	Correlation	1
How much does Y change with X, other things equal?	Linear regression	2
Does it hold within countries over time?	Panel regression	3
Is a country in the high or the low group?	Classification	4
Can many indicators be summarised in a few?	PCA	5
Which countries look alike?	Clustering	6
Some values are missing. What now?	Missing values	7

01

Correlation

Do two indicators move together across countries?

One number for "do they move together?"

The correlation coefficient r goes from −1 to +1. Close to +1: when one indicator is high, the other tends to be high. Close to −1: one high, the other low. Close to 0: no linear pattern.

1

Always look at the scatter plot first

One dot per country. The picture shows outliers and curves that r cannot.

2

r measures straight-line patterns

A strong curved relationship can have a small r.

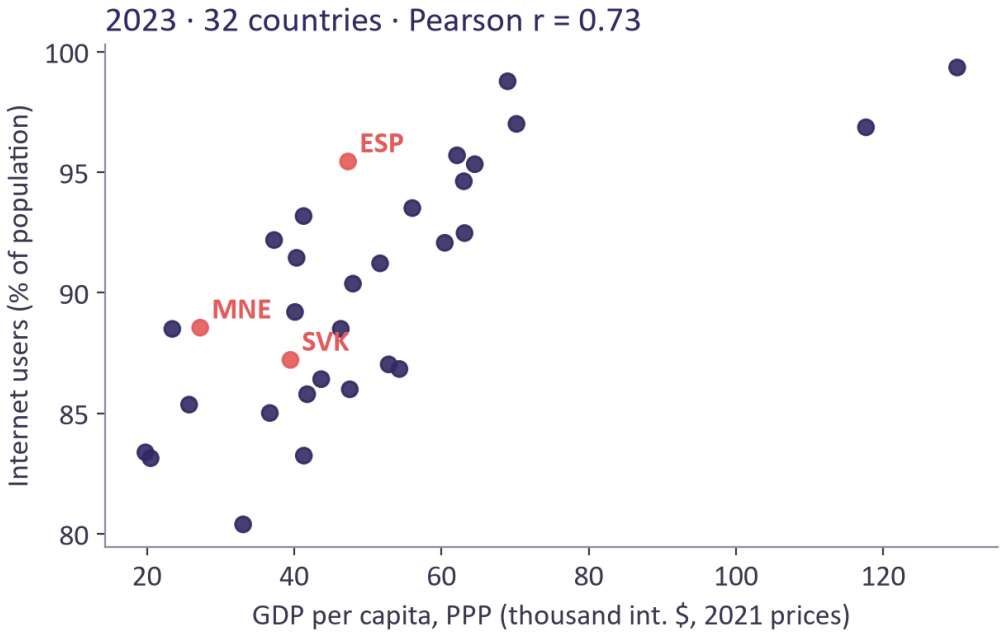
3

Same year for every country

Comparing 2015 in one country with 2023 in another mixes time and country.

CORRELATION · EXAMPLE

Richer countries use the Internet more



r = 0.73

Internet users vs GDP per capita, 2023

32

countries (EU aggregate excluded)

Strong

positive association; without Luxembourg and Ireland, far to the right, r is still 0.74

Source: World Bank (IT.NET.USER.ZS, NY.GDP.PCAP.PP.KD). Spain, Slovakia and Montenegro in coral.

What r = 0.73 tells you, and what it does not

How to read it

- Countries with higher income per person tend to have a larger share of Internet users.
- Spain is the country furthest above the line: about 6 points more users than expected for its income. Montenegro is slightly above, Slovakia slightly below.
- Rough guide: |r| below 0.3 weak, 0.3 to 0.7 moderate, above 0.7 strong.

Watch out

- It does not say that income causes Internet use (or the reverse).
- Two very rich countries can pull r up or down: check without them.
- Use percentages or per-capita values, never totals, when countries differ in size.

Correlation in your tool

Power BI · menus, no code

- Scatter chart: X = GDP per capita, Y = Internet users, Values = country.
- Filter year = 2023.
- Analytics pane > Trend line.
- New quick measure > Correlation coefficient.

Gretl · menus, no code

- Sample > Restrict, based on criterion: year == 2023.
- Select the two variables.
- View > Correlation matrix.
- View > Graph specified vars > X-Y scatter.

WITH CODE · ONLY IF YOUR COURSE USES PYTHON OR R

Python

```
s = d[d.year == 2023]
print(s[["IT_NET_USER_ZS", "NY_GDP_PCAP_PP_KD"]].corr())
```

R

```
s <- subset(d, year == 2023)
cor(s$IT_NET_USER_ZS, s$NY_GDP_PCAP_PP_KD, use = "complete.obs")
```

02

Linear regression

How much does one indicator change when another changes, keeping the rest constant?

A line through the cloud of countries

$Y = b0 + b1 \cdot X1 + b2 \cdot X2 + \text{error}$. Each coefficient b is the average change in Y when that X increases by one unit and the other X stay the same.

1 **Coefficient (b)**

Size and sign of the association, in the units of Y per unit of X.

2 **p-value**

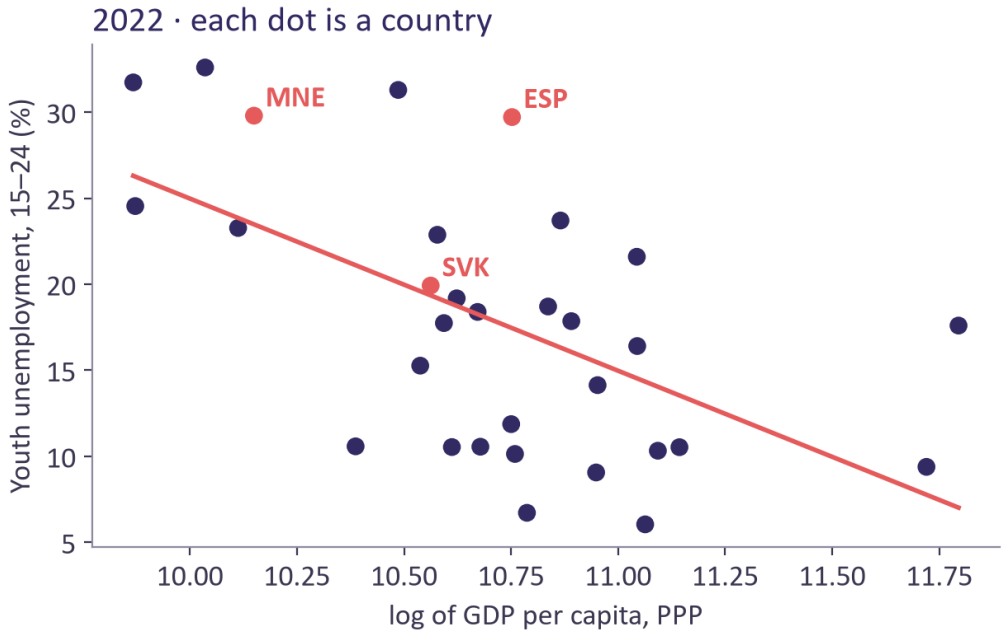
Below 0.05: the pattern is unlikely to be pure chance. It says nothing about how large or important it is.

3 **R^2**

Share of the differences in Y between countries that the model reproduces (0 to 1).

REGRESSION · EXAMPLE

Youth unemployment, education and income



-10.0

coefficient of log GDP per capita (p = 0.001)

0.006

coefficient of tertiary enrolment (p = 0.909)

$R^2 = 0.32$

31 countries, 2022

Model: youth unemployment = b0 + b1·tertiary enrolment + b2·log(GDP per capita). The line in the chart uses income only.

Reading the table in plain English

How to read it

- Income: with GDP per capita 10% higher, youth unemployment is about 1.0 points lower, for the same tertiary enrolment.
- Tertiary enrolment: no association once income is taken into account (p = 0.909).
- The model reproduces about 32% of the differences between countries; the rest depends on other things (labour laws, training systems…).
- With a log X: coefficient × 0.095 ≈ change in Y for a 10% higher X.

Watch out

- Association, not effect: income could stand for many other things.
- 31 countries: add two or three variables at most.
- Look for outliers in the scatter before you trust the line.
- Do not try models until one is significant: decide the model from your question.

Regression in your tool

Power BI · menus, no code

- One X: scatter chart + Analytics pane > Trend line shows the fitted line.
- Two or more X: Power BI has no regression table. Use Gretl (no code) or a Python/R visual.

Gretl · menus, no code

- Sample > Restrict: year == 2022.
- Select GDP per capita, then Add > Logs of selected variables.
- Model > Ordinary Least Squares.
- Dependent: SL_UEM_1524_ZS. Regressors: SE_TER_ENRR and l_NY_GDP_PCAP_PP_KD.

WITH CODE · ONLY IF YOUR COURSE USES PYTHON OR R

Python

```
s = d[d.year == 2022]
f = "SL_UEM_1524_ZS ~ SE_TER_ENRR + log_gdp"
m = smf.ols(f, data=s).fit()
print(m.summary())
```

R

```
s <- subset(d, year == 2022)
m <- lm(SL_UEM_1524_ZS ~ SE_TER_ENRR + log_gdp, data = s)
summary(m)
```

03

Panel data and time trends

The same countries followed over the years: compare each country with itself.

Between countries or within a country?

A simple regression on all country-years mixes two things: differences between countries and changes within each country. Country fixed effects keep only the changes within each country.

1

Country fixed effects

Everything stable about a country (history, institutions, climate) is taken out.

2

Time trend

Captures what changes for everybody over the years.

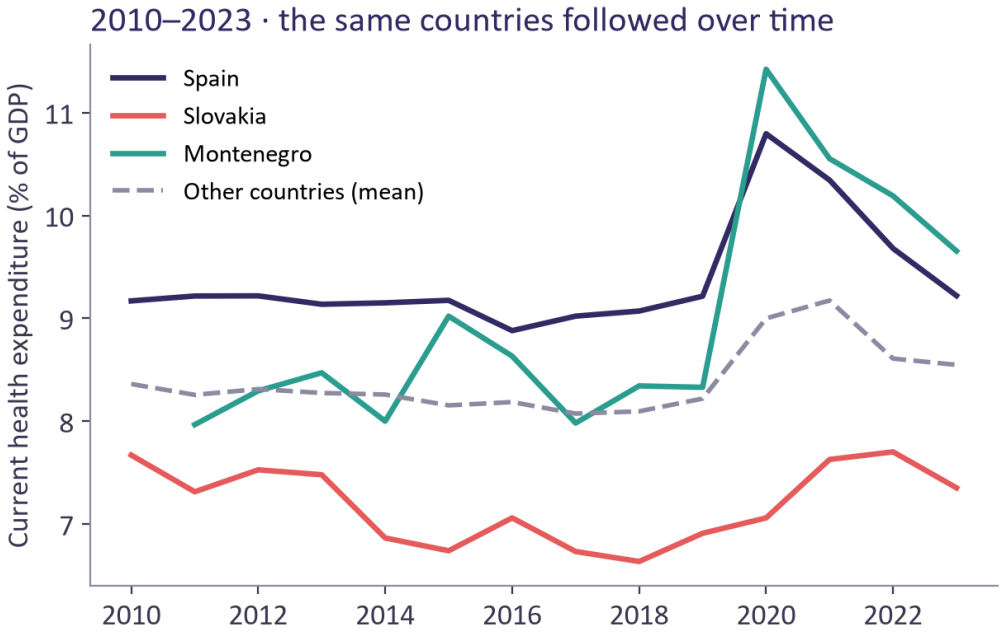
3

Clustered standard errors

Years of the same country are not independent: cluster by country.

PANEL · EXAMPLE

Ageing and health expenditure, 2010–2023



+0.25

all country-years together ($p < 0.001$)

-0.10

within countries, with fixed effects and trend ($p = 0.276$)

+0.08

points of GDP per year, common trend ($p = 0.056$)

447 observations, 32 countries. Coefficient: change in health spending (% of GDP) per extra point of population aged 65+.

PANEL · READING

Why the two answers differ

How to read it

- Between countries: older societies spend more on health.
- Within countries: in these 14 years, ageing faster did not go with spending more, once the common trend is removed.
- Both are true; they answer different questions. Say which one your question is about.

Watch out

- 2020 (COVID-19) moves every country: look at the chart, consider year dummies.
- With ordinary standard errors p = 0.08; clustered by country p ≈ 0.27. Always cluster.
- Small differences between software (0.27 vs 0.28) are normal.

Panel regression in your tool

Power BI · menus, no code

- Line chart: X = year, Y = health expenditure, Legend = country: shows the panel.
- The model itself: use Gretl (no code) or a Python/R visual.

Gretl · menus, no code

- Data > Dataset structure > Panel (unit iso3, time year).
- Sample > Restrict: year <= 2023.
- Add > Define new variable: trend = year - 2010.
- Model > Panel > Fixed or random effects: choose Fixed effects, tick Robust standard errors.

WITH CODE · ONLY IF YOUR COURSE USES PYTHON OR R

Python

```
y, x = "SH_XPD_CHEX_GD_ZS", "SP_POP_65UP_TO_ZS"
p = d[d.year.between(2010, 2023)].dropna(subset=[y, x]).copy()
p["trend"] = p.year - 2010
f = f"{y} ~ {x} + trend + C(iso3)"
m = smf.ols(f, data=p).fit(cov_type="cluster",
    cov_kwds={"groups": pd.factorize(p.iso3)[0]})
print(m.params[[x, "trend"]], m.pvalues[[x, "trend"]])
```

R

```
library(fixest)   # install.packages("fixest") once
p <- subset(d, year >= 2010 & year <= 2023)
p$trend <- p$year - 2010
m <- feols(SH_XPD_CHEX_GD_ZS ~ SP_POP_65UP_TO_ZS + trend | iso3,
           data = p, cluster = ~iso3)
coeftable(m)
```

04

Classification

Predict the group of a country: high or low?

From a number to a probability

Logistic regression gives each country a probability (0 to 1) of being in the "high" group. Above 0.5 we classify it as high. Then we count how many we got right.

1. **Define the groups first**

   For example above or below the median of the indicator.

2. **Confusion matrix**

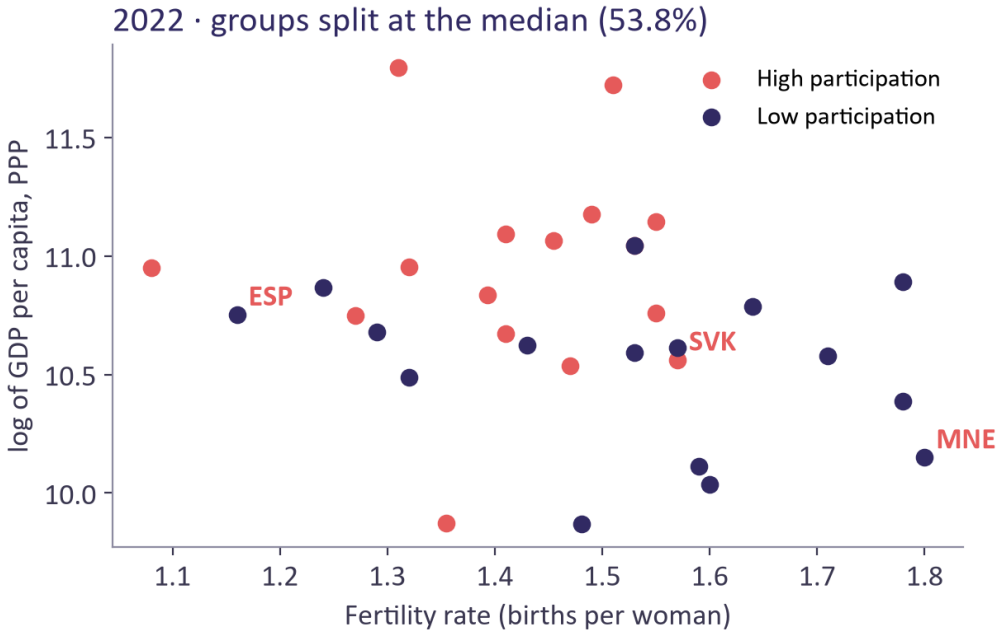
   A 2×2 table: real group against predicted group.

3. **Accuracy**

   Share of countries classified correctly. Compare it with 50%, the result of guessing.

CLASSIFICATION · EXAMPLE

High or low female participation?



75%

of 32 countries classified correctly (24 of 32)

53.8%

median participation: the line between high and low

p = 0.042

income helps to classify; fertility does not (p = 0.217)

Logistic regression: high participation ~ fertility rate + log GDP per capita, 2022.

Reading the confusion matrix

How to read it

- Real low, predicted low: 12. Real low, predicted high: 4.
- Real high, predicted high: 12. Real high, predicted low: 4.
- Spain is classified high but is low; Slovakia is classified low but is high. Wrong cases are interesting: explain them.

Watch out

- 75% measured on the same countries used to fit the model is optimistic. With so few countries, use leave-one-out cross-validation if you can.
- Splitting at the median throws information away: say why you do it.

Classification in your tool

Power BI · menus, no code

- Key influencers visual: Analyze = high/low group, Explain by = fertility and GDP per capita. Good for exploring.
- The model and the confusion matrix: Gretl (no code) or a Python/R visual.

Gretl · menus, no code

- Sample > Restrict: year == 2022.
- Add > Define new variable: high = SL_TLF_CACT_FE_ZS > 53.8
- Model > Limited dependent variable > Logit > Binary.
- Dependent: high. Regressors: SP_DYN_TFRT_IN, l_NY_GDP_PCAP_PP_KD. The output reports the cases correctly predicted.

WITH CODE · ONLY IF YOUR COURSE USES PYTHON OR R

Python

```
s = d[d.year == 2022].copy()
fem = s.SL_TLF_CACT_FE_ZS
s["high"] = (fem > fem.median()).astype(int)
m = smf.logit("high ~ SP_DYN_TFRT_IN + log_gdp", data=s).fit()
s["pred"] = (m.predict(s) > 0.5).astype(int)
print(pd.crosstab(s.high, s.pred))
```

R

```
s <- subset(d, year == 2022)
fem <- s$SL_TLF_CACT_FE_ZS
s$high <- as.integer(fem > median(fem))
m <- glm(high ~ SP_DYN_TFRT_IN + log_gdp, family = binomial, data = s)
s$pred <- as.integer(predict(m, type = "response") > 0.5)
table(real = s$high, predicted = s$pred)
```

05

Principal component analysis

Summarise many related indicators in two or three new ones.

Fewer variables, almost the same information

Many indicators say similar things (richer countries have more Internet users and live longer). PCA builds new variables, the components, that collect what the indicators share.

1 **Standardise first**

Otherwise the indicator with the biggest numbers dominates.

2 **Variance explained**

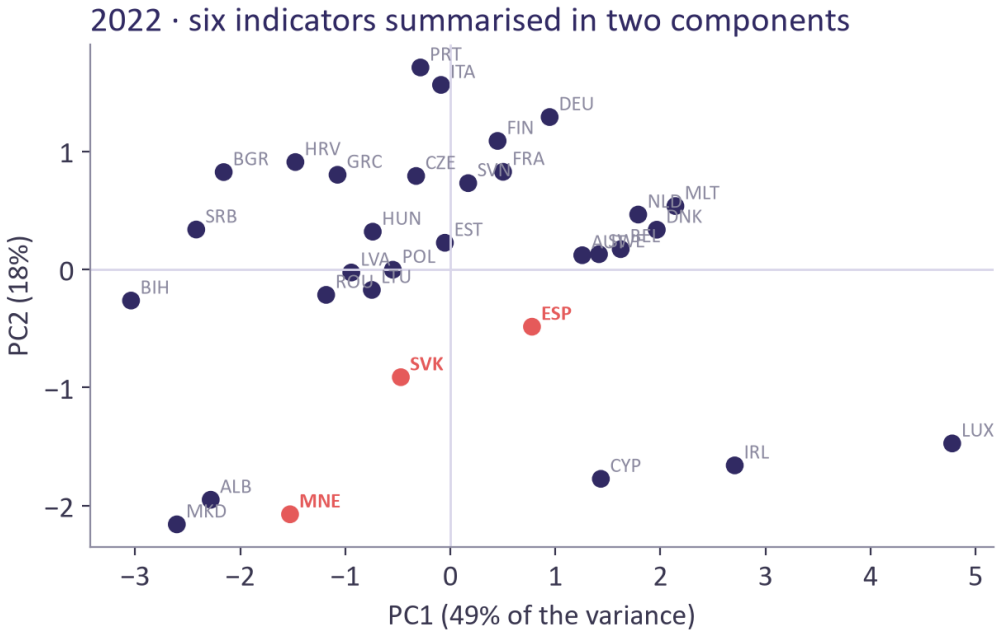
How much information each component keeps.

3 **Loadings**

How much each indicator weighs in a component. They tell you what the component means.

PCA · EXAMPLE

Six indicators, two components



49%

of the variance in component 1: "development"

18%

in component 2: "ageing"

67%

kept by the two together

2022, standardised: GDP per capita, services, Internet users, population 65+, unemployment, life expectancy.

Naming the components from the loadings

How to read it

- Component 1: GDP per capita 0.53, Internet 0.5, life expectancy 0.45, services 0.4, unemployment -0.29. High = more developed.
- Component 2: population 65+ 0.87, unemployment -0.41. High = older population.
- In the chart, Montenegro is low on both; Spain is near the centre on development.

Watch out

- The sign of a component is arbitrary: two programs can flip it.
- Naming components is your interpretation: show the loadings.
- Use the components afterwards in clustering or regression if you need fewer variables.

PCA in your tool

Power BI · menus, no code

- Power BI has no PCA. Use Gretl (no code), or a Python/R visual with the code below.

Gretl · menus, no code

- Sample > Restrict: year == 2022.
- Select the six variables.
- View > Principal components.
- Gretl uses the correlation matrix by default, so the variables are standardised.

WITH CODE · ONLY IF YOUR COURSE USES PYTHON OR R

Python

```
cols = ["NY_GDP_PCAP_PP_KD", "NV_SRV_TOTL_ZS", "IT_NET_USER_ZS",
        "SP_POP_65UP_TO_ZS", "SL_UEM_TOTL_ZS", "SP_DYN_LE00_IN"]
s = d[d.year == 2022].dropna(subset=cols)
z = StandardScaler().fit_transform(s[cols])
pca = PCA().fit(z)
print(pca.explained_variance_ratio_.round(3))
print(pd.DataFrame(pca.components_[:2].T, index=cols).round(2))
```

R

```
cols <- c("NY_GDP_PCAP_PP_KD", "NV_SRV_TOTL_ZS", "IT_NET_USER_ZS",
          "SP_POP_65UP_TO_ZS", "SL_UEM_TOTL_ZS", "SP_DYN_LE00_IN")
s <- na.omit(subset(d, year == 2022)[, cols])
pc <- prcomp(s, scale. = TRUE)
summary(pc)
round(pc$rotation[, 1:2], 2)
```

06

Clustering

Which countries look alike?

Groups of similar countries

k-means puts each country in one of k groups so that countries in the same group are as close as possible to the group centre. You choose k.

1 **Standardise first**

So that every indicator counts the same.

2 **Choose k**

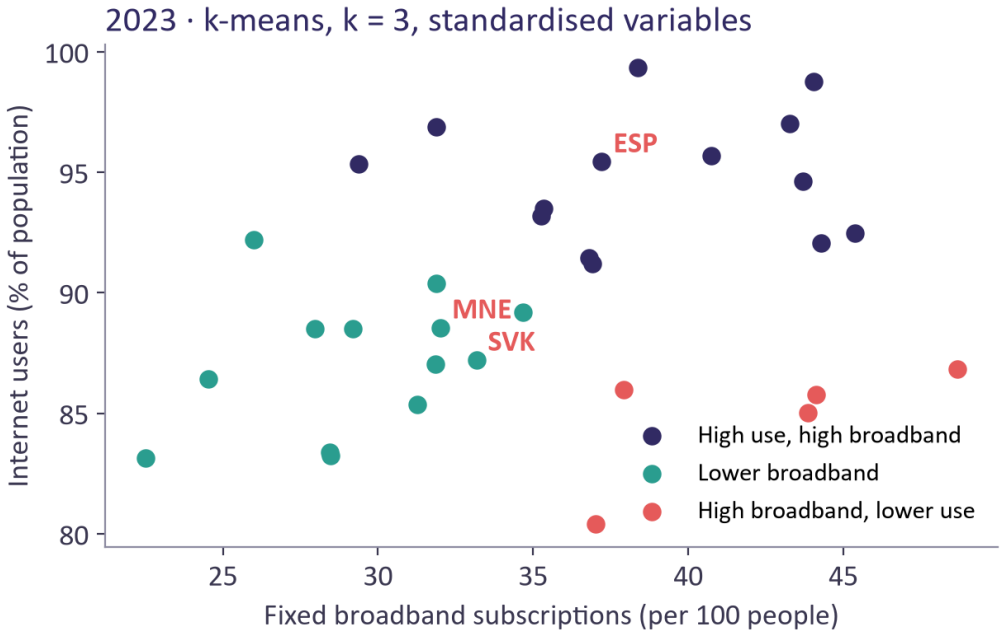
Silhouette score (higher is better) and whether the groups make sense.

3 **Describe the groups**

By their centres (averages), not by the cluster number.

CLUSTERING · EXAMPLE

Three digital profiles in Europe, 2023



14

High use, high broadband: 94.8% use the Internet, 38.8 broadband per 100

13

Lower broadband: 87.2% use the Internet, 29.4 broadband per 100

5

High broadband, lower use: 84.8% use the Internet, 42.3 broadband per 100

k-means, k = 3 (silhouette 0.432, the highest between 2 and 6), Internet users and fixed broadband, standardised.

Where do our three countries fall?

How to read it

- Spain: High use, high broadband. Slovakia and Montenegro: Lower broadband.
- High broadband, lower use: BGR, CZE, FRA, GRC, PRT: many broadband lines but fewer users.
- Silhouette by k: 2 → 0.397, 3 → 0.432, 4 → 0.381, 5 → 0.38, 6 → 0.384.

Watch out

- k-means always finds groups, even if there are none: check the silhouette and the chart.
- Fix the random seed and use many starts, or the groups change each time.
- Check if the groups are stable in another year.

Clustering in your tool

Power BI · menus, no code

- Scatter chart: X = broadband, Y = Internet users, Values = country.
- Filter year = 2023.
- More options (…) > Automatically find clusters.
- Set the number of clusters to 3.

Gretl · menus, no code

- Gretl has no built-in k-means.
- Use Power BI (no code), or Python or R.

WITH CODE · ONLY IF YOUR COURSE USES PYTHON OR R

Python

```
cols = ["IT_NET_USER_ZS", "IT_NET_BBND_P2"]
s = d[d.year == 2023].dropna(subset=cols).copy()
z = StandardScaler().fit_transform(s[cols])
km = KMeans(n_clusters=3, n_init=25, random_state=2026)
s["cluster"] = km.fit_predict(z)
print(s.groupby("cluster")[cols].mean().round(1))
print(s.groupby("cluster").iso3.apply(list))
```

R

```
cols <- c("iso3", "IT_NET_USER_ZS", "IT_NET_BBND_P2")
s <- na.omit(subset(d, year == 2023)[, cols])
set.seed(2026)
k <- kmeans(scale(s[, -1]), centers = 3, nstart = 25)
split(s$iso3, k$cluster)
```

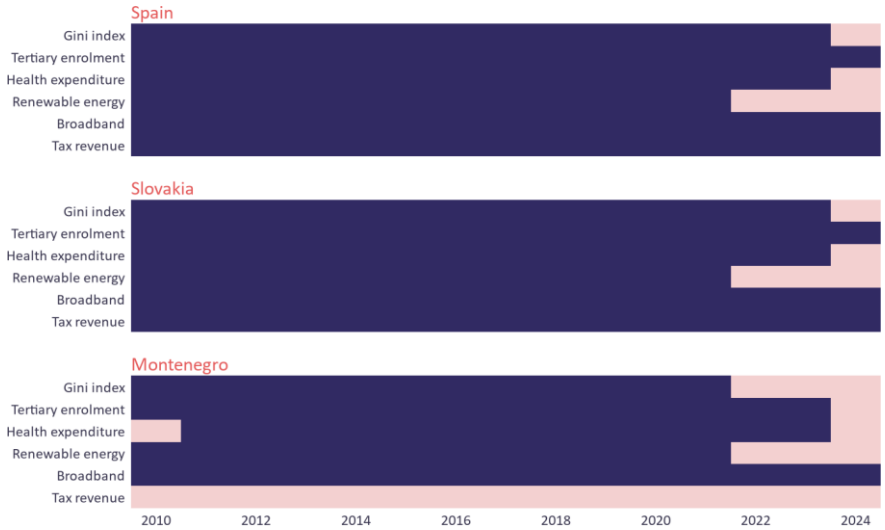
07

Missing values

Gaps are normal. Hiding them is not.

MISSING VALUES

Four honest options



Dark = available, pink = missing, 2010–2024

1. Change year or indicator
2. Leave the country out, and say so (n = …)
3. Interpolate only between two known years, and mark it
4. Add a documented source (Eurostat, MONSTAT…)

Find gaps: Power BI > Transform data > View > Column quality. Gretl: Variable > Summary statistics (number of missing values).

FROM RESULTS TO THE DASHBOARD

Six rules for charts people understand

1. One message per chart; write it as the title ("Richer countries use the Internet more").
2. Label axes with units, year and source.
3. Highlight Spain, Slovakia and Montenegro; keep the others grey.
4. Bars for comparing countries, lines for time, scatter for two indicators.
5. No 3D, no pie charts with many slices.
6. State the limits on the dashboard: n, years, gaps, association not causation.

All the code, ready to run

On the website (Resources): methods.py (Python), methods.R (R) and methods.inp (Gretl). They reproduce every number in this module. Put them in the same folder as coil_indicators_gretl.csv.

Stuck? Ask your home teacher, or bring the problem to the mid-point clinic (week 5).